

## Integrating Differential Equations

A **differential equation** is an equation relating a function and one or more of its derivatives. The solutions to a differential equation are all functions that satisfy the equation.

Differential Equation

General Solution

$$\frac{dy}{dx} = 2x$$

$$y = x^2 + c$$

$$y'' + 9y = 0$$

$$y = A \cos 3x + B \sin 3x$$

In the second case notice that:

$$y = A \cos 3x + B \sin 3x$$

$$y' = -3A \sin 3x + 3B \cos 3x$$

$$y'' = -9A \cos 3x - 9B \sin 3x$$

Thus we have:

$$y'' + 9y = (-9A \cos 3x - 9B \sin 3x) + 9(A \cos 3x + B \sin 3x) = 0.$$

Hence  $y = A \cos 3x + B \sin 3x$  is a solution to the differential equation

$$y'' + 9y = 0.$$

$y'' + 9y = 0$  is called a 2nd order differential equation because the second derivative of  $y$  appears in the equation.

**Initial Value Problem:** The solution to a differential equation given an initial condition or conditions.

Ex. Solve  $\frac{dy}{dx} = 4x^3$ ; with the initial condition that  $y(1) = 3$ .

General Solution:  $y = x^4 + c$

Initial Condition:  $y(1) = 1^4 + c = 3$

$$\Rightarrow c = 2$$

$y = x^4 + 2$  is called the **Particular Solution**.

$y = x^4 + c$  is called the **General Solution**.

If we have  $\frac{dy}{dx} = g(x)$  or  $\frac{d^2y}{dx^2} = g(x)$  we can try integrating both sides.

Ex. Solve  $\frac{dy}{dx} = x^2\sqrt{x^3 + 1}$ ;  $y(2) = 3$ .

**General Solution:**

$$y = \int x^2 (x^3 + 1)^{\frac{1}{2}} dx$$

Let  $u = x^3 + 1$

$$du = 3x^2 dx$$

$$\frac{1}{3} du = x^2 dx$$

$$\int x^2 (x^3 + 1)^{\frac{1}{2}} dx = \frac{1}{3} \int u^{\frac{1}{2}} du = \frac{1}{3} \left( \frac{2}{3} u^{\frac{3}{2}} \right) + c$$

$$y = \frac{2}{9} (x^3 + 1)^{\frac{3}{2}} + c$$

**Particular Solution:**

$$y(2) = \frac{2}{9}(2^3 + 1)^{\frac{3}{2}} + c = 3$$

$$\frac{2}{9}(9)^{\frac{3}{2}} + c = 3$$

$$\frac{2}{9}(27) + c = 3$$

$$6 + c = 3$$

$$c = -3$$

$$\boxed{y = \frac{2}{9}(x^3 + 1)^{\frac{3}{2}} - 3}$$

Ex. Find the General Solution of  $\frac{d^2y}{dx^2} = 6x + \sin x$ .

$$\frac{dy}{dx} = \int 6x + \sin x \, dx = 3x^2 - \cos x + c_1$$

$$y = \int (3x^2 - \cos x + c_1) \, dx = x^3 - \sin x + c_1x + c_2.$$

### **Velocity and Acceleration**

When a particle moves in a line and the position is given by  $x(t)$  then,

$$v(t) = \frac{dx}{dt} = \text{velocity}$$

$$a(t) = \frac{d^2x}{dt^2} = \text{acceleration} = \frac{dv}{dt}.$$